INCH-POUND

A-A-59749
February 25, 2003

COMMERCIAL ITEM DESCRIPTION

COOKER, STEAM, PRESSURELESS
(NAVAL SHIPBOARD)

The General Services Administration has authorized the use of this commercial item description (CID), for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers pressureless steam cookers.

1.2 Intended use. The steam cooker is intended for use in Navy shipboard galleys.

2. CLASSIFICATION

2.1 Classification. The steam cooker shall conform to the following types:

Type 1: Vacuum steamer
Type 2: Pressureless steamer

Beneficial comments (recommendations, additions, deletions) and any data which may be of use in improving this document should be addressed to: US. Army Soldier and Biological Chemical Command, Natick Soldier Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3. SALIENT CHARACTERISTICS

3.1 Design and construction. The steam cookers shall consist of a water tight insulated cooking cavity. Cooking shall be accomplished by direct action of steam as specified under type designation (see 2.1). The steamer shall have a manual valve that drains water into a removable 4 inch deep pan located under the steamer, or equipped with a suitable connection to the drain. All connections shall be designed so that the unit may be connected while maintaining a flush rear or side surface. Each steamer type shall be fabricated from stainless steel with corrosion-resistant hardware and fittings in accordance with the manufacturer's standard practice. All threaded parts shall conform to ANSI B1.1.

Each cooker type shall provide the following features:

- Single cooking cavity
- Switch that cuts electrical power to the heater when water level is low
- Rail guides capable of being removed or adjusted without the use of tools.
- Control panel shall be accessible for serviceability
- A 60 minute(minimum) timer which will give an audible signal at the end of a cooking cycle

3.2 Dimensions. Each steamer type shall have the following maximum outside dimensions: 24 inches wide by 31 inches deep by 33 1/2 inches high (excluding legs).

3.3 Stackability. Each steamer type unit shall have the capability to be stacked one upon another without binding or wobbling, or use a rack to accomplish this requirement.

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3.4 Steamer performance. The cooking chamber shall be capable of generating steam within 15 minutes or less after turn on. The steamer shall use a maximum of one gallon of water per hour. The steam cookers shall be capable of cooking at temperatures between 150^{0}F – 212^{0}F for type 1, and 212^{0}F for type 2 when tested as specified in 5.2.1.

3.5 Steamer capacity. Each cooking compartment shall have the capacity to accommodate the following pan sizes and quantities: six steam table pans(12 inches by 20 inches by 2-1/2 inches high), or four steam table pans(12 inches by 20 inches by 4 inches high), or three steam table pans (12 inches by 20 inches by 6 inches high). Pans shall not contact the interior oven wall surfaces.

3.6 Doors. Steamer door shall be stainless steel, with replaceable gasket, and provision made to prevent condensate from dripping on the deck. The door shall not require more than 16 lbs of force to unlatch during operation, when tested as specified in 5.2.2.

3.7 Door latch. Steamer door shall be equipped with a secondary exterior mechanical latch (in addition to the main door latch) that shall allow the door to open approximately 1/2 inch to release trapped steam inside cooking cavity. Each door shall be provided with a hold-open latch which holds the door open in a 90 degree, or greater open position.

3.8 Mounting. The steamer unit shall be provided with four type 300 series stainless steel round legs, each a minimum 1 inch in diameter, 4 inches in length, for securing the unit to a dresser, countertop, or stand.

3.9 Inclined operation and structural integrity (shipboard requirement, see 7.2). The steamers shall operate satisfactorily, with no leakage of fluid or product, when inclined at an 15 degrees each side of the vertical in each of two vertical planes at right angles to each other, when used for surface ships. For submarine use, the angle of inclination shall be 30 degrees. During testing, the door shall remain closed to prevent the loss of contents, and the steamer shall not rack by more than 1/8-inch.

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3.10 Environmental suitability . Steamer shall be capable of withstanding ship's vibration in accordance with MIL-STD-167/1, type 1 environmental vibration without malfunction.

3.11 Temperature control. Each steamer shall be equipped with an automatic, or adjustable thermostat to provide temperature control of the cooking compartment. The thermostat shall be capable of maintaining temperature range of 150 ^{0}F to 212 ^{0}F for type 1, and 212 ^{0}F for type 2.

3.12 Heating elements. All heating elements shall be sealed and protected to preclude damage by moisture.

3.13 Surface temperature. Each steamer unit shall be capable of operating at 212 ^{0}F with no external contact surface exceeding a temperature of 140 ^{0}F.

3.14 Electrical requirements. Unless otherwise specified, steamers for shipboard use shall be compatible with 440 VAC, 60 Hz, 3 phase, 3-wire, underground of 115 VAC, 60 Hz, single phase, alternating current systems for shipboard, as specified in MIL-STD-1399/300.

3.15 Electrical wiring. Electrical wiring shall be arranged, supported and marked to allow identification with provided wiring diagram for each unit located on access panel.

3.16 Controls. All controls, switches, and indicator lights shall be readily accessible, easily operable, and shall be located in a waterproof control panel on the front of the unit.

3.17 Accessibility. The access panel and components for each unit shall be physically and visually accessible for operation and for normal maintenance with tools, test equipment, and replacement parts. No special tools shall be required for access or parts replacement.

3.18 Physical size limitations. The units shall pass or shall disassemble without any need of cutting and/or modification, so that it can pass through a shipboard access measuring 26 inches wide by 66 inches high.

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3.19 Label plates. The unit shall be provided with a data nameplate and an instruction plate, both attached to the front doors of the unit. They shall be readily visible during normal operating use and shall not adversely affect the life and utility of the unit.

3.19.1 Data nameplate. The data nameplate shall contain the manufacturer's name, model, serial number, date manufactured, and other information needed to uniquely identify the unit.

3.19.2 Instruction plate, and operation manual. The unit shall be provided with an instruction plate affixed to the front door, and an operations manual for start-up, operation, shut down, and cleaning.

3.20 High voltage labels. A label reading "Danger-High Voltage" shall be affixed to the outer case assembly, on or adjacent to each service access cover near one of the fasteners securing the cover. In addition, a warning label in accordance with UL 969 shall be placed near the high voltage components inside the equipment. This label shall include, but not be limited to the following texts:

- Danger-High Voltage
- Power supply must be disconnected before servicing
- Access covers must be in place before use
- Service should be performed by authorized personnel only

3.21 Standard compliance. The unit shall be in accordance with and certified to meet the applicable requirements of NSF 4 and UL 197.

3.22 Workmanship. The finished units shall be free of cracks, dents, or scratches. All burrs, slivers, splinters, sharp edges, dirt, grease, and foreign matter shall be removed.

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3.23 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of IEEE/ASTM SI 10, and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENT

4.1 Contractor recovered material. The offeror/contractor is encouraged to use recovered materials to maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The product provided shall meet salient characteristics of this CID, conform to the producer’s own drawings, specifications, standards, and quality assurance practices, and/or be the same product that has successfully been delivered to the Government for shipboard use on a previous procurement. The Government reserves the right to require proof of such compliance to all salient characteristics.

5.2 TESTING

5.2.1 Steaming mode performance test. Prior to testing, the steamer cooking cavity shall be preheated for 15 minutes. With the steamer in the steam mode, two-dozen large eggs shall be placed in a 2 inch perforated pan, and loaded into the cooking cavity and held for 13 minutes. Three eggs selected at random shall be opened with in one minute to determine compliance with 3.4. All sample eggs shall exhibit a solid yoke.

5.2.2 Door pull test. The oven door shall not require more than 16 pounds of force to open when operating at 212 ^{0}F to determine compliance with 3.6.

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6. PACKAGING

6.1 Packaging. Preservation, packing, and marking shall be as specified in the contract or purchase order.

7. NOTES

7.1 Operation and repair manual, inspection and testing reports. Each unit shall be supplied with an inspection and testing report; and an operation and repair manual.

7.2 Ordering data. Purchasers should include the following information in procurement documents.

a. Title, number and date of this document.
b. Type required (see 2.1).
c. Packaging requirements(see 6.1).

7.3 Sources of document.

7.3.1 Sources of government documents.

MILITARY STANDARDS:

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally excited)

MIL-STD-1399 - Interface Standard for Shipboard Systems Section 300A Electric Power, Alternating Current

(Unless otherwise indicated, copies of the above standards are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

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7.3.2 Sources for non-government association documents. Copies of referenced non-government documents may be obtained from the following organizations:

AMERICAN NATIONAL STANDARDS INSTITUTE:

ANSI B1.1 – Unified Inch Screw Threads(UN & UNR Thread Form)

(Unless otherwise indicated, copies of the above standard is available from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

NSF INTERNATIONAL:

NSF 4 – Commercial Cooking, Rethermalization and Powered Hot Food Holding and Transport Equipment

UNDERWRITERS LABORATORIES INC.:

UL 197 – Commercial Electric Cooking Appliances
UL 969 – Marking and Labeling Systems

(Unless otherwise indicated, copies of the above standards are available from the Underwriters Laboratories Inc., 333 Pfingsten Road, Northbrook, IL 60062).

IEEE/ASTM STANDARD

SI 10 – Standard for Use of the International System of Units (SI): The Modern Metric System

(Unless otherwise indicated, copies of the above standard is available from the Institute of Electrical and Electronics Engineers, Inc., 345 East 47th Street, New York, NY 10017, or the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428).

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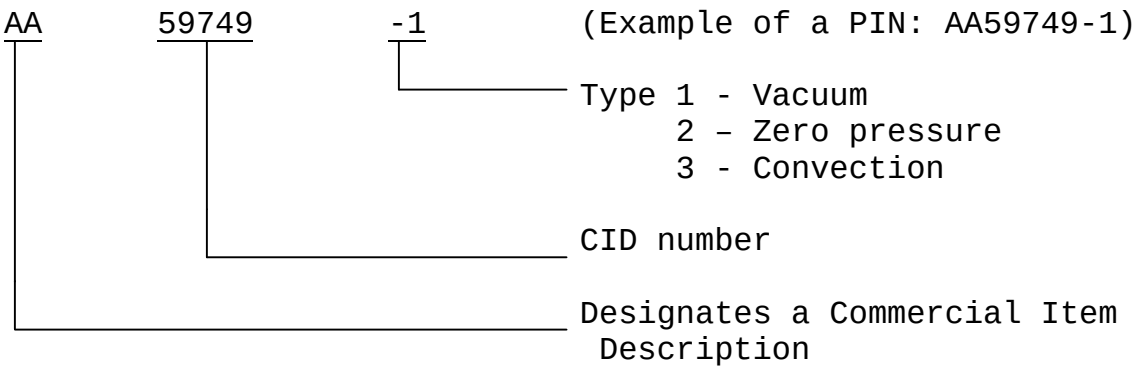
7.4 Sources of supply. A suppliers who are known to manufacture a cooker, steam, pressureless are listed below; however, competition is not limited to these companies.

AccuTemp Products Inc
8415 North Clinton Park
Fort Wayne, IN 46825

Market Forge Industries Inc.
35 Garvey Street
Everett, MA 02149-4403

3 Rivers Equipment Inc.
Building 5, Suite C
10347 Dawson Creek Blvd
Ft Wayne, IN 46825

7.5 Part identification number(PIN). The PINs to be used for items required by this CID are created as follows:



MILITARY INTERESTS:

Custodians
Army – GL
Navy – SA

Review Activities
Army –MD1
Navy – SH
DLA – SS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS/7FLE

Preparing Activity:

Army - GL

DoD Project (7310-0938)

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